

# RENUKA PRASAD PATWARI

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## Education

**Masters of Science in Software Engineering** | San Jose State University **Aug 2025 - May 2027**  
Coursework: Software Systems Engineering, Enterprise Software Platforms, Machine Learning CGPA: 3.62/4

**Bachelors of Engineering in Computer Engineering** | University of Mumbai **Aug 2020 - May 2024**  
Coursework: Data Structures & algorithms, Natural Language Processing (NLP), Artificial Intelligence CGPA: 3.96/4

## Technical Skills

**Programming Languages:** Python, Java, TypeScript, JavaScript, SQL

**Cloud:** AWS (Bedrock, Lambda, CloudFront, DynamoDB, S3, EC2, IAM), GCP (GKE, Vertex AI, BigQuery, Cloud Run, Firestore)

**Backend & Databases:** Spring Boot, Microservices, REST APIs, Kafka, PostgreSQL, MySQL, Redis

**Libraries & Frameworks:** PyTorch, Scikit-learn, NumPy, Pandas, Flask, Django, FastAPI, React

**AI Tools & Skills:** Claude Code, Codex, Cursor, Gemini, Github Copilot, Prompt Engineering

**Tools & Testing:** Pytest, JUnit 5, Mockito, Docker, Swagger, Postman, GitHub, Prometheus, Grafana

## Work Experience

**Assistant Software Engineer** | Stellantis **Sept 2024 – Jul 2025**

- Delivered secure policy creation, search, and renewal workflows by building object-oriented Java Spring Boot microservices for a Vehicle Insurance System using JDBC, Eureka, and H2.
- Increased unit test coverage from 3% to 90% using JUnit 5 and Mockito, identifying and resolving critical test gaps to strengthen overall code quality.

**Machine Learning Intern** | Indian Institute of Technology, Bombay (IITB) **Sep 2023 – Mar 2024**

- Improved model selection accuracy by evaluating 3 AI models (PyTesseract, DocTr, Google Vision API) across multilingual scripts to identify the best-performing OCR solution.
- Accelerated deployment reliability by containerizing the application with Docker and documenting REST APIs with Swagger and Postman, enabling reproducible and consistent model iteration.

## Projects

**DevLens**

*Google ADK, MCP, BigQuery, FastAPI, Next.js*

- Consolidated developer productivity signals across 3 data sources by building a Gemini-powered ADK agent on Vertex AI that surfaces sprint health, PR review lag, and workload imbalances from GitHub, Jira, and Google Calendar via Fivetran.
- Automated cross-tool developer workflows across 2 MCP servers by integrating Fivetran for data sync and GitHub for human-in-the-loop issue creation, streaming all MCP activity live across 4 real-time SSE channels to a Next.js dashboard.
- Deployed full-stack on Google Cloud Run with a FastAPI async backend, 5 BigQuery derived metric views, and Firestore for persisting chat sessions, alerts, and issue proposals across requests.

**CodeGuardian**

*GraphRAG, ChromaDB, Gemini, FastAPI, Neo4j*

- Reduced vulnerability detection time to 0.43s per scan by building a FastAPI web service and CLI tool that integrates into developer workflows, covering 13 OWASP and CWE vulnerability categories.
- Achieved 95.74% F1 accuracy by architecting a Planner-Executor-Hybrid Critic multi-agent pipeline combining GraphRAG retrieval with LLM override logic, validated on a 120-case benchmark measuring precision, recall, FPR, and latency.

**VeriFi**

*Kafka, Redis, PostgreSQL, Docker, Kubernetes, Grafana*

- Sustained 155.8 TPS with 12ms p95 latency for real-time fraud detection by building a distributed Kafka pipeline for high-volume transaction ingestion and processing, validated with Locust.
- Enhanced edge-case fraud decisioning with an XGBoost model achieving 0.8295 ROC AUC by combining ML predictions with 8 rule-based fraud signals.
- Deployed a 5-microservice architecture on Kubernetes (GKE), configuring HPA, liveness and readiness probes, ConfigMaps, and Secrets, and instrumented with Prometheus and Grafana for real-time metrics tracking and alerting.

## Hackathons & Competitions

**SJSU AWS Cloudathon** | 2nd Place out of 300+ Participants **May 2026**

- Secured 2nd place out of 300+ participants by resolving 14 AWS security challenges in 4 hours, covering IAM policy fixes, CloudFront origin protection, Lambda automation, RDS/S3 encryption, and EC2 security group misconfigurations.

**NVIDIA GTC Agents for Impact Hackathon** | PaperGap **March 2026**

- Engineered a 5-phase AI agent pipeline that analyzed 250M+ research papers and automatically surfaced the top 3 underexplored research gaps per topic, reducing manual literature review effort for researchers.